

Fuzzy-skin texture and the peak electrical output of a small vertical-axis wind turbine rotor

Wind-tunnel comparison of four blade sets, 1 October 2026

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Data URI_VAWT_Texture_Data_2026-10-01.zip (accompanies this report)

Summary

Four blade sets for one three-bladed vertical-axis rotor were compared in the URI wind tunnel on 1 October 2026. The sets differ in fuzzy-skin thickness: none (Plain) and 0.050 mm, 0.101 mm and 0.202 mm (FS 0.05, FS 0.10 and FS 0.20; earlier labels Ra 20, Ra 40 and Ra 80). Each rotor was swept twice through 14 wind speeds from 10.2 to 38.0 m/s with an automated constant-current load, giving 112 set-point values of peak electrical power $P_{\max}$ (Appendix A). Table 1 summarises the result.

Table 1. Peak electrical power relative to the Plain rotor, averaged over the 14 wind speeds, with measured surface roughness.

Rotor	Fuzzy skin (mm)	Pa (μm)	Ra (μm)	Change ^a (%)	$P_{\max}$ at 38.0 m/s (W)
Plain	none	19.0	12.7	reference	3.67
FS 0.05	0.050	13.9	9.4	+1.8 [−0.6, +4.2]	3.78
FS 0.10	0.101	13.2	8.2	+12.8 [+10.1, +15.5]	4.30
FS 0.20	0.202	28.7	10.3	+16.2 [+13.5, +19.0]	4.53

^aGeometric mean over wind speeds of the ratio to the Plain rotor, both runs, with Tukey simultaneous 95% intervals from run-to-run repeatability on one mounting ($\sigma = 0.58\%$). These are intervals for the four rotors as mounted, not for the effect of texture (Section 7). $P_{\max}$ at 38.0 m/s: geometric mean of the two runs. Pa, Ra: profilometer (Section 2.3); FS 0.20 mean of two fields.

Findings.

1. The FS 0.10 and FS 0.20 rotors produced +12.8% and +16.2% more peak electrical power than the Plain rotor. FS 0.05 (+1.8% [−0.6, +4.2]) is not resolved from Plain. Five of the six pairwise differences are resolved (Tukey simultaneous 95% intervals exclude zero). The smallest, FS 0.20 over FS 0.10 (+3.0% [+0.6, +5.5]), is marginal: with the parabolic-fit estimator its lower limit is +0.03%, and mounting variation above about 0.3% would remove it.
2. The advantage depends on wind speed (rotor $\times$ set point interaction, $F_{39,52} = 9.2$, $p < 0.001$). The power coefficient of FS 0.10 and FS 0.20 rises more steeply, and at lower wind speed, than Plain's, so the gains are largest at 21–25 m/s.
3. The gain appears in the generator's extrapolated open-circuit voltage (+6.4% and +8.5% for FS 0.10 and FS 0.20), which indicates faster rotation at light load; averaged over wind speed, the

apparent source resistance does not differ from Plain's. The rig did not log rotor speed; tip-speed ratio and $C_P(\lambda)$ will be added from T. Kang's rotor-speed record.

4. Measured roughness does not rank the rotors by power. R_a is highest on Plain (12.7 μm , from its 0.2 mm layer lines), and it ranks FS 0.05 above FS 0.10, the reverse of their power. FS 0.05 and FS 0.10 have similar P_a (13.9 and 13.2 μm) but differ by +10.8% in power; one field per set does not confirm that their realised texture differs.

Limits. Each texture level is one print, mounted once, and the rotors were run in order of increasing texture. The results compare these four rotors as mounted; attributing the differences to texture needs replicate prints and mountings and an interleaved run order (Section 9). Section 5 gives how much mounting variation, and Section 7 how much session drift, the differences can absorb.

Responses to earlier requests. Blade identity (Prof. Jahangiri): Table 2. Fault-testing plan (Prof. Jeong): Section 9.1. Comparison with the earlier laboratory tests (Prof. Jeong): outside the scope of this report (Section 1).

1 Objective and scope

The objective is to compare the peak electrical power of a small vertical-axis wind turbine (VAWT) rotor fitted with four blade sets that differ in printed fuzzy-skin texture, and to characterise the printed surfaces. Fuzzy skin is a slicer feature that, in each layer, moves points on the outer-wall path, about one point distance apart, by a random distance of up to $\pm$ a set thickness normal to the wall.

This report covers one test session, 1 October 2026: four blade sets, two runs each, and a same-day profilometer scan of each set. Earlier laboratory (June–July) and rig (August–September) tests are outside its scope. A fifth blade set also run that day is excluded because its print settings could not be confirmed. Data and code accompany the report (Appendix E).

2 Test articles

2.1 Rotor

All four rotors use one geometry: three straight, untwisted blades attached at $R = 0.1016\text{ m}$ from the axis, with span $H = 0.2451\text{ m}$. The nominal swept area $A = 2RH = 0.0498\text{ m}^2$ is used for $C_{P,\text{el}}$ only; the outer swept radius is larger and was not documented. The blade section is a 183° circular arc, close to a semicircular cup, not an airfoil: 48 mm chord, 24.5 mm depth, 1.79 mm wall and square edges. Over the test the chord Reynolds number based on wind speed is 3.2×10^4 to 1.2×10^5 (air at 20 °C).

2.2 Blade sets

Table 2 lists the sets. The controlled variable is the fuzzy-skin thickness; the earlier labels Ra 20/40/80 are names, not measured values, and are not used here.

The slicer project (`turbine_default.3mf`, Bambu Studio, saved 17 August) holds one blade per plate, $24.5 \times 48.0 \times 245.1\text{ mm}$, for the FS 0.10 and FS 0.20 settings and for one set not reported. It specifies a Bambu Lab P1S with a 0.2 mm nozzle, 0.10 mm layers and PLA, with fuzzy skin painted on both faces of each blade (96.5% of the surface area; the square edges and end caps are plain) at

Table 2. Blade sets. Layer period: dominant spanwise wavelength of the measured surface (Appendix C).

Rotor	Fuzzy skin	Earlier label	Layer period	Print record
Plain	none	baseline	0.20 mm	No slicer record; printer, nozzle and material unknown. One blade broke at the end of run 2; $P_{\max}$ at 1800 rpm matches run 1 within 0.1%.
FS 0.05	0.050 mm	Ra 20	0.10 mm	No slicer record; thickness from the 1 October run log.
FS 0.10	0.101 mm	Ra 40	0.10 mm	Slicer project, plate “40 RA”. One blade carried a small sensor magnet in an earlier test; whether it was still fitted on 1 October was not recorded.
FS 0.20	0.202 mm	Ra 80	—	Slicer project, plate “80 RA”. No distinct layer period; the fuzz dominates.

a 0.2 mm point distance. It does not record which prints were made from it, and the 1 October run logs record only the fuzzy-skin thickness, so the material and nozzle of the four sets are unconfirmed.

2.3 Surface measurement

Each set was scanned on 1 October before the runs (13:07–13:22) with a Keyence VR-6000 optical profilometer (VR-6100 head, $\times 160$): one field of about 2.0×1.4 mm on one blade per set (two fields for FS 0.20) at $1.85 \mu\text{m}$ per pixel; the location on the blade was not recorded (Fig. 1). Profiles run spanwise, across the layer lines, and each is levelled. Pa is the arithmetic mean deviation of the levelled profile. Ra is that of the roughness profile, the levelled profile minus its Gaussian mean line (ISO 16610-21 [1], $\lambda_c = 0.25$ mm), over a 1.13–1.17 mm evaluation length (Appendix C). Each value is the mean over the pixel columns of a field; adjacent columns are correlated, so the field is the sampling unit.

- The Plain surface is not smooth: its layer lines form a regular 0.20 mm ridge pattern and give it the highest Ra of the four ($12.7 \mu\text{m}$).
- FS 0.05 and FS 0.10 have similar Pa (13.9 and $13.2 \mu\text{m}$) and Ra (9.4 and $8.2 \mu\text{m}$), on the same 0.10 mm layer period. Pa roughly doubles from FS 0.10 to FS 0.20 (ratio 2.2), as the setting does, but not from FS 0.05 to FS 0.10 (ratio 0.95); from one field each, the scans do not confirm that these two sets differ in realised texture.
- FS 0.20 has about twice their Pa ($28.7 \mu\text{m}$). Its two fields differ by $4.8 \mu\text{m}$ in Pa (17%) and $2.0 \mu\text{m}$ in Ra, so one field is a coarse sample.

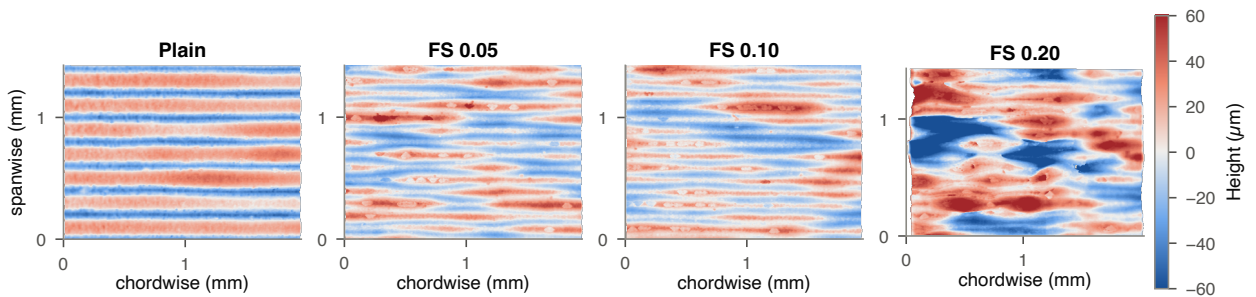


Figure 1. Profilometer height maps, one field per blade set, each column levelled. Layer lines run horizontally (chordwise); profiles for Pa and Ra run vertically (spanwise). Colour clipped at $\pm 60 \mu\text{m}$; FS 0.20 exceeds this over 8% of its area.

3 Facility and instrumentation

3.1 Wind speed

The tunnel fan is driven by an ABB ACS550 variable-frequency drive, commanded in rpm by a Portenta Machine Control over Modbus (Fig. 2). Wind speed v is obtained from the commanded fan speed N with the tunnel’s calibration (Test 1, 13 February 2026), a least-squares line through eleven points (Appendix D):

$$v = 0.02132 N - 0.424 \quad (\text{m/s; } N \text{ in rpm}).$$

Only eight points were measured, from 0 to 700 rpm (up to 14.0 m/s). The calibration file marks the other three, at 1400–2400 rpm, as possibly read from the trend line of the original record, and the Test 1 instrument is not documented. The eleven set points above 700 rpm (16.6–38.0 m/s) therefore rely on air speed remaining linear in fan speed. At 500–700 rpm the adopted line is 2.8–4.6% above the measured speeds. A line through the measured points alone gives 36.0 m/s at 1800 rpm, 5% below the adopted value. This affects absolute wind speeds and $C_{P,\text{el}}$, not the comparison between rotors, which are compared at identical commanded fan speed.

The median fan-motor current at each set point agreed across the eight runs within 0.1 A, the drive’s reading resolution (13.6 A at 1800 rpm), so the fan operating point did not change measurably during the session. Air temperature and pressure were not recorded. No blockage correction is applied; because the rotors share one geometry, blockage acts on them alike to first order.

3.2 Electrical measurement

The rotor shaft drives a three-phase generator. Its output passes through a three-phase bridge rectifier to a Chroma 63004-150-60 DC electronic load operated in constant-current mode. A Python host sets the load current and records the terminal voltage V and current I reported by the load at the end of each dwell; electrical power is $P = VI$. Every comparison in this report is between quantities measured with the same instrument at the same settings, so instrument calibration errors largely cancel. The rig did not log rotor speed; T. Kang recorded it separately, and every dwell carries a Unix time stamp for joining that record.

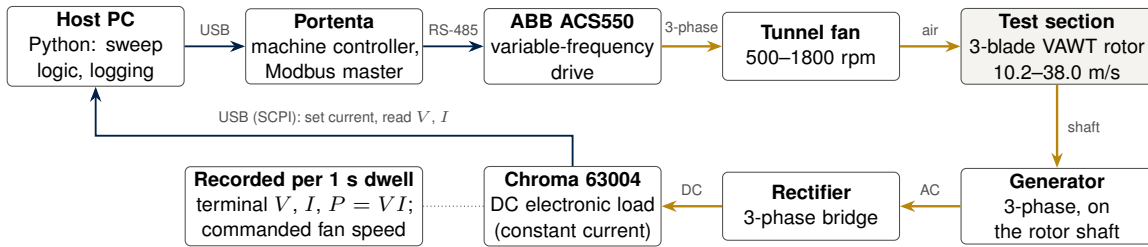


Figure 2. Measurement and control chain.

4 Test procedure

4.1 Sweep protocol

Each run (sweep) visits 14 fan set points from 500 to 1800 rpm in steps of 100 rpm. At each set point the host waits until the fan speed is within the larger of 10 rpm and 2% of the command, and

then until the generator’s terminal voltage has changed by no more than 1% (and no more than 0.02 V where that is larger) over each of three successive 0.5 s intervals (at least 2 s; each phase at most 20 s). The load then raises its current demand in steps of 20 mA at 1800 rpm (scaled with v^2 at lower speeds, minimum 2 mA), dwelling 1 s per step, and stops after two consecutive dwells at or below 80% of the running maximum. The resulting load ladder of 13–30 dwells traces the power–current curve through its maximum (Fig. 3). A run takes about 8 minutes. All runs used the same automated protocol (protocol identifier 94bed28333f7).

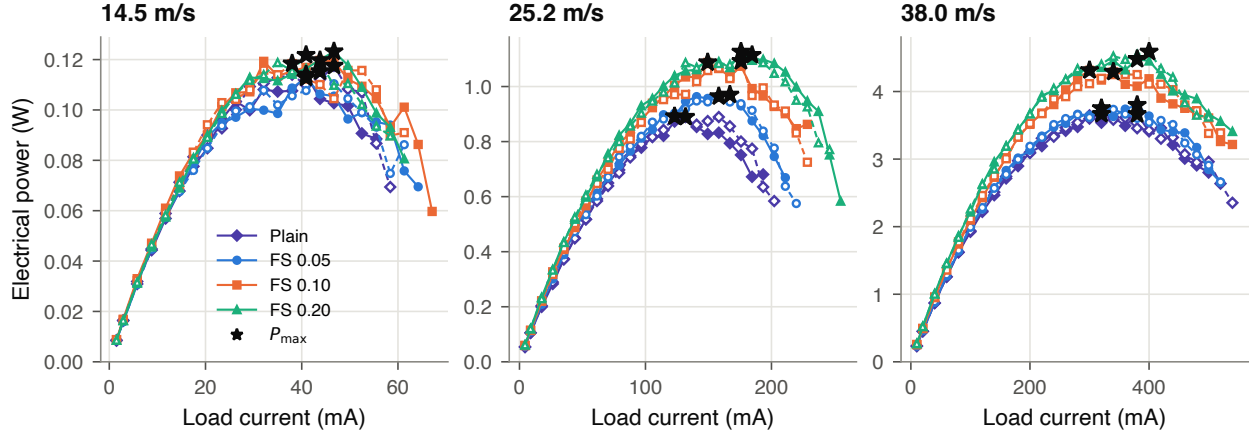


Figure 3. Measured load ladders at three wind speeds, all eight runs (solid: run 1; dashed, hollow: run 2). Stars mark $P_{\max}$, the largest dwell power.

4.2 Session

Each rotor was mounted once and swept twice without being removed (Table 3), in the order Plain, FS 0.05, FS 0.10, FS 0.20, from 15:50 to 18:24. The two Plain runs started 60 minutes apart; the other pairs ran back to back.

Table 3. Runs of 1 October 2026 (local time of the first and last dwell). Change: run mean relative to the Plain rotor (Section 5).

Rotor	Run	Time	Change (%)	Data file stem
Plain	1	15:50–15:57	−0.4	v1_smooth_20261001
	2	16:50–16:58	+0.4	v1_smooth_repeat_20261001
FS 0.05	1	17:18–17:25	+2.3	v1_Ra20_20261001
	2	17:27–17:34	+1.3	v1_Ra20_repeat_20261001
FS 0.10	1	17:42–17:50	+12.4	v1_Ra40_20261001
	2	17:51–17:59	+13.1	v1_Ra40_repeat_20261001
FS 0.20	1	18:07–18:15	+15.7	v1_Ra80_20261001
	2	18:16–18:24	+16.8	v1_Ra80_repeat_20261001

4.3 Data quality

Of the 112 ladders, 110 stopped on the power roll-off. The other two stopped on the load's voltage-floor cut-out after the peak, with the last dwell already below 80% of the maximum: Plain run 1 at 500 rpm (49%) and FS 0.05 run 2 at 600 rpm (57%). The peak is therefore inside the ladder in all 112. Peak powers recomputed from the dwell records equal the values logged by the rig exactly.

5 Data reduction and statistical analysis

Peak power. $P_{\max}$ is the largest dwell power in the ladder. As a check, a least-squares parabola is fitted to the maximum and up to two dwells either side (the rig software's rule); in ten ladders the fit is not concave or its vertex falls outside those dwells, and the maximum is used. The two estimators give the same conclusions except where noted (Table 4). Each dwell lasts 1 s and rotor speed was not logged, so it is not shown that the rotor reaches equilibrium within a dwell; at 500 rpm it was still accelerating after the settling check (Table 6). $P_{\max}$ is therefore the peak of this ramp protocol.

Rotor comparison. The analysis uses $\ln P_{\max}$, so differences are ratios. For each run, y is the mean over the 14 set points of $\ln(P_{\max}/P_{\text{ref}})$, where P_{ref} is the geometric mean of the two Plain runs at that set point; each wind speed carries equal weight. The pooled run-to-run standard deviation of y is $\sigma = 0.58\%$ (4 degrees of freedom). The rotor effect is tested with a one-way analysis of variance (ANOVA) on the eight run means, and all six pairs are compared with Tukey's honestly significant difference procedure, which gives simultaneous 95% intervals (family-wise error rate 5%; studentized range $q_{0.05;4,4} = 5.76$) [2]. A difference is called resolved when its interval excludes zero. With two runs per rotor the p -values rest on a normal model and are given only down to 0.001; an exact permutation test of the rotor effect on the eight run means gives $p = 1/105$, the smallest attainable. Because each rotor's two runs were consecutive, neither test separates a rotor effect from drift (Section 7).

Mounting and drift. The two runs of a rotor share its print and mounting, so these intervals measure the repeatability of the measurement on one mounting. Mounting-to-mounting variation τ , the standard deviation of a rotor's mean power between mountings, was not measured. A difference stays resolved by the same Tukey criterion for τ up to 2.0% (FS 0.10 vs Plain), 2.6% (FS 0.20 vs Plain), 1.7% (FS 0.10 vs FS 0.05), 2.3% (FS 0.20 vs FS 0.05) and 0.3% (FS 0.20 vs FS 0.10). Because the rotors were run in order of increasing texture, the run means are also refitted with a linear time term (Section 7).

Dependence on wind speed. A two-way ANOVA with runs nested in rotors tests the rotor $\times$ set point interaction against the residual (run $\times$ set point within rotor), that is, whether the ratio between rotors changes with wind speed by more than the point-to-point scatter of repeated runs (Appendix B). The residual scatter differs between set points, so this F test is approximate; it is repeated without the two set points with the largest scatter.

Source of the gain. At each set point a straight line $V = V_{\text{oc}} - IR_{\text{int}}$, a Thévenin equivalent, is fitted to the dwells before any cut-out ($r^2 \geq 0.965$). V_{oc} is its zero-current intercept, an extrapolated open-circuit voltage; for a permanent-magnet generator it rises in proportion to shaft speed, less a rectifier drop. R_{int} is an apparent source resistance. It includes the winding, rectifier and wiring and the rotor slowing as the load rises; the latter is large, since a fixed resistance cannot explain the fall from $91\ \Omega$ at $10.2\ \text{m/s}$ to $36\ \Omega$ at $38.0\ \text{m/s}$ for Plain. $V_{\text{oc}}^2/4R_{\text{int}}$ is 0.91–1.04 times $P_{\max}$ over all ladders. Both quantities are compared between rotors as $P_{\max}$ is.

Power coefficient. The electrical power coefficient is $C_{P,el} = P_{max}/(\frac{1}{2}\rho Av^3)$ with standard air, $\rho = 1.204 \text{ kg/m}^3$. It includes generator and rectifier losses and is not the aerodynamic power coefficient. Because $C_{P,el}$ is only 0.09–0.28%, drivetrain losses may be a large share of the power the rotor extracts, and a fractional change in aerodynamic power can appear magnified in P_{max} . The changes reported here are changes in system output.

6 Results

6.1 Power curves

Peak electrical power (rotor means of the two runs) rises from 0.030–0.034 W at 10.2 m/s to 3.67–4.53 W at 38.0 m/s (Fig. 4a). The electrical power coefficient rises with wind speed for every rotor, from 0.09% to 0.28% (Fig. 4b). With the measured-points-only calibration, $C_{P,el}$ at 38.0 m/s would be 17% higher (relative).

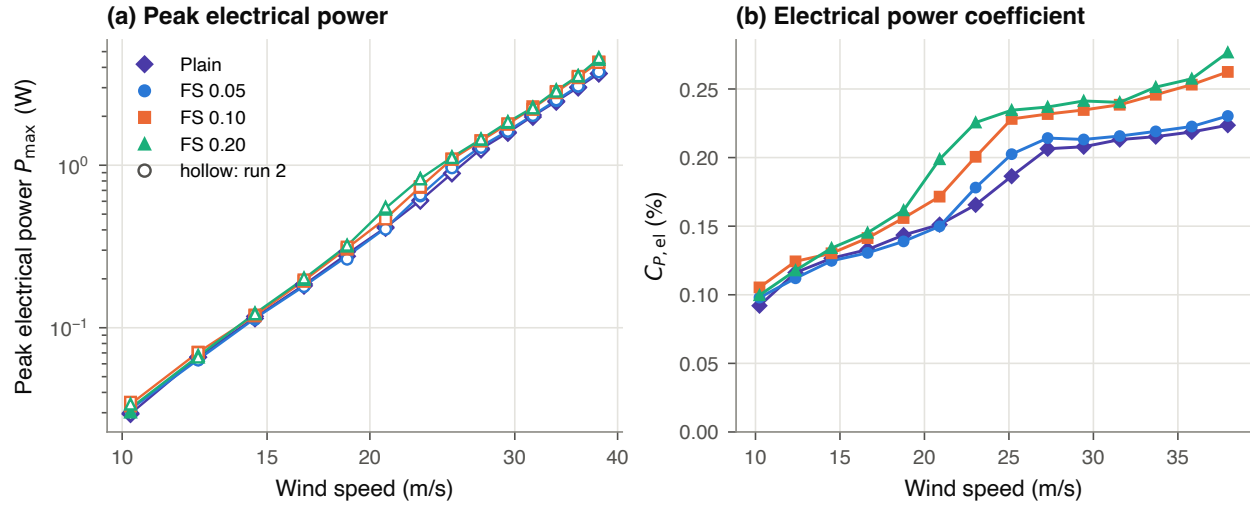


Figure 4. (a) Peak electrical power of every run, logarithmic axes (filled: run 1; hollow: run 2); lines join the geometric means of the two runs. (b) Electrical power coefficient, standard air density.

6.2 Rotor comparison

The rotor effect is large compared with run-to-run variation ($F_{3,4} = 324$, $p < 0.001$; permutation $p = 1/105$). The ratio of the two run means is within 1% for every rotor (Fig. 5), and their scatter is not detectably larger than the point-to-point scatter within runs ($F_{4,52} = 1.39$, $p = 0.25$). Five of the six pairwise differences are resolved (Table 4); FS 0.05 vs Plain is not. FS 0.20 over FS 0.10 is the smallest resolved difference; with the parabolic-fit estimator its lower limit is +0.03% ($p = 0.048$).

6.3 Dependence on wind speed

The difference between rotors is not uniform in wind speed (Fig. 6). The rotor $\times$ set point interaction is large compared with the scatter of repeated runs ($F_{39,52} = 9.2$, $p < 0.001$). It remains so among the three textured rotors alone ($F_{26,39} = 7.6$, $p < 0.001$) and without the two set points with the largest scatter, 500 and 900 rpm ($F_{33,44} = 23.7$, $p < 0.001$). The steepest segment of $\ln P_{max}$ against $\ln v$, resolved to the 2.1 m/s set-point spacing, lies at 20 m/s for FS 0.20 and 22 m/s for FS 0.10 in

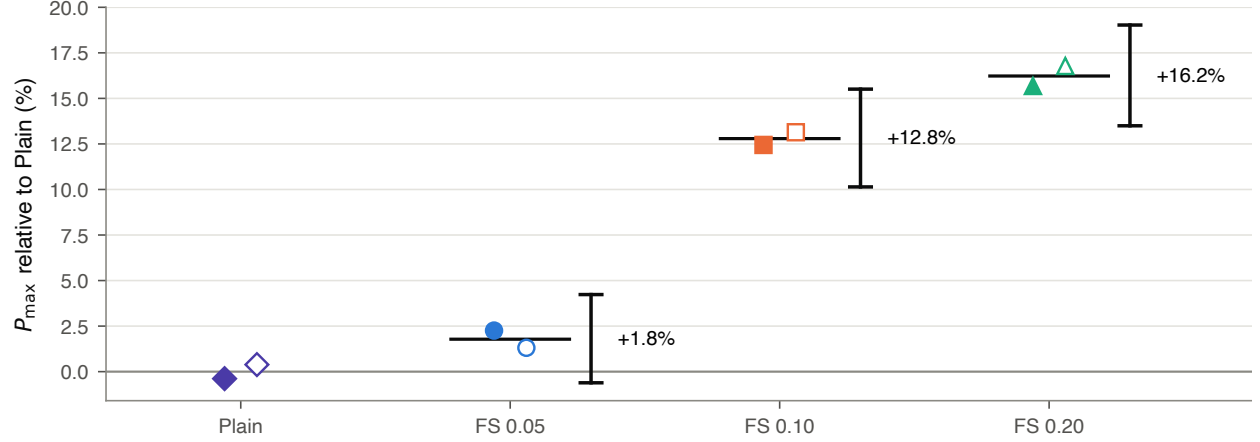


Figure 5. Each run’s mean $P_{\max}$ relative to the Plain rotor (filled: run 1; hollow: run 2). Bars: rotor mean with Tukey simultaneous 95% interval for the difference from Plain.

Table 4. Pairwise comparisons of $P_{\max}$, geometric mean over wind speeds: Tukey simultaneous 95% intervals and adjusted p -values. Last column: the same for the parabolic-fit estimator.

Comparison	Change (%)	95% CI (%)	p	Parabolic fit (%)
FS 0.05 vs Plain	+1.8	[−0.6, +4.2]	$p = 0.12$	+1.6 [−1.2, +4.4]
FS 0.10 vs Plain	+12.8	[+10.1, +15.5]	$p < 0.001$	+13.0 [+10.0, +16.2]
FS 0.20 vs Plain	+16.2	[+13.5, +19.0]	$p < 0.001$	+16.2 [+13.1, +19.5]
FS 0.10 vs FS 0.05	+10.8	[+8.2, +13.5]	$p < 0.001$	+11.3 [+8.3, +14.4]
FS 0.20 vs FS 0.05	+14.2	[+11.5, +16.9]	$p < 0.001$	+14.4 [+11.3, +17.6]
FS 0.20 vs FS 0.10	+3.0	[+0.6, +5.5]	$p = 0.023$	+2.8 [+0.03, +5.7]

both runs, at 22–24 m/s for FS 0.05 and at 24–26 m/s for Plain; its slope is 4.9–5.0 for FS 0.20 and 4.4 for Plain. FS 0.10 and FS 0.20 rise below both Plain runs; FS 0.05 overlaps Plain. The gains are largest between 21 and 25 m/s: FS 0.20 exceeds +25% only there, peaking at +36.2% at 23 m/s, and FS 0.10 peaks at +22.4% at 25 m/s. Quoted wind speeds carry the calibration uncertainty of Section 3.1; the ratios do not.

6.4 Source of the gain

Averaged over wind speed, the textured rotors’ extrapolated open-circuit voltage is higher than Plain’s, and their source resistance does not differ from it (Table 5, Fig. 7). The V_{oc} changes alone correspond to power changes of +2.4%, +13.1% and +17.7%, against +1.8%, +12.8% and +16.2% measured. R_{int} does vary between rotors with wind speed ($F_{39,52} = 5.7$, $p < 0.001$): for FS 0.20 it is −7.1% at 21 m/s and +9.2% at 38 m/s relative to Plain. It is below Plain’s from 15 to 25 m/s and above it from 27 m/s, a pattern common to all three textured rotors and so possibly from the Plain reference. Where the gains are largest, the lower R_{int} supplies up to 43% of the gain in $\ln P_{\max}$. A higher open-circuit voltage at the same fan speed indicates faster rotation at light load. Whether this comes from more aerodynamic torque or less parasitic torque cannot be determined from these data.

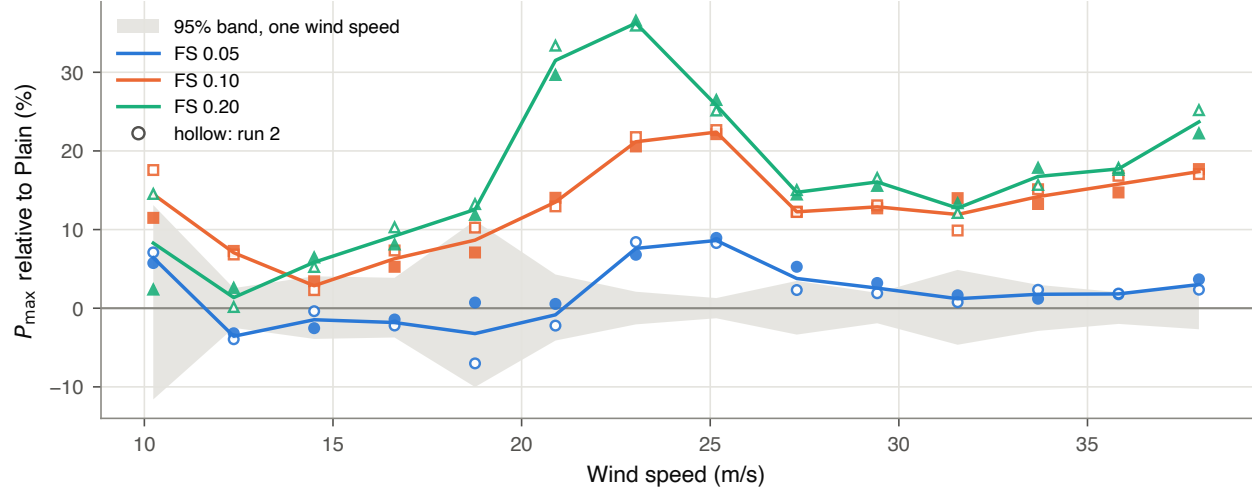


Figure 6. $P_{\max}$ of each textured rotor relative to the Plain rotor at each wind speed. Markers: individual runs (filled: run 1; hollow: run 2); lines: geometric mean of the two runs. Band: 95% interval of a single-wind-speed difference from run-to-run scatter. All curves share the Plain reference, so features common to all three may come from it.

Table 5. Thévenin parameters relative to Plain, mean over wind speeds, with Tukey simultaneous 95% intervals; electrical power coefficient at 38.0 m/s, standard air.

Rotor	V_{oc} (%)	R_{int} (%)	$C_{P,el}$ (%)
Plain	reference	reference	0.22
FS 0.05	+1.2 [+0.2, +2.2]	−0.1 [−1.5, +1.4]	0.23
FS 0.10	+6.4 [+5.3, +7.4]	−0.1 [−1.5, +1.4]	0.26
FS 0.20	+8.5 [+7.4, +9.6]	+0.2 [−1.3, +1.6]	0.28

6.5 Surface roughness and power

The measured surface parameters do not order the rotors as their power does (Table 1). R_a is highest on Plain, from its layer lines, and ranks FS 0.05 above FS 0.10; the textured sets span 8.2–10.3 μm , about the 2.0 μm by which the two FS 0.20 fields differ. P_a ranks FS 0.20 highest and Plain second, and does not separate FS 0.05 from FS 0.10, although their power differs by +10.8%: either the realised texture of these two sets does not follow the setting, or one field per set does not resolve it. The power ranking matches the nominal fuzzy-skin thickness; with one print per level this is an association (Section 7).

7 Uncertainty and limitations

Table 6 lists the sources of uncertainty. Two are not covered by the intervals.

- **Print and mounting.** Each texture level is one print mounted once, so print-to-print and mounting variation are confounded with texture; Section 5 gives the mounting variation each difference could absorb. The Plain set also differs in layer height (0.2 mm against about 0.1 mm), and its printer, nozzle and material are unknown. FS 0.05 and FS 0.10 share a 0.10 mm layer

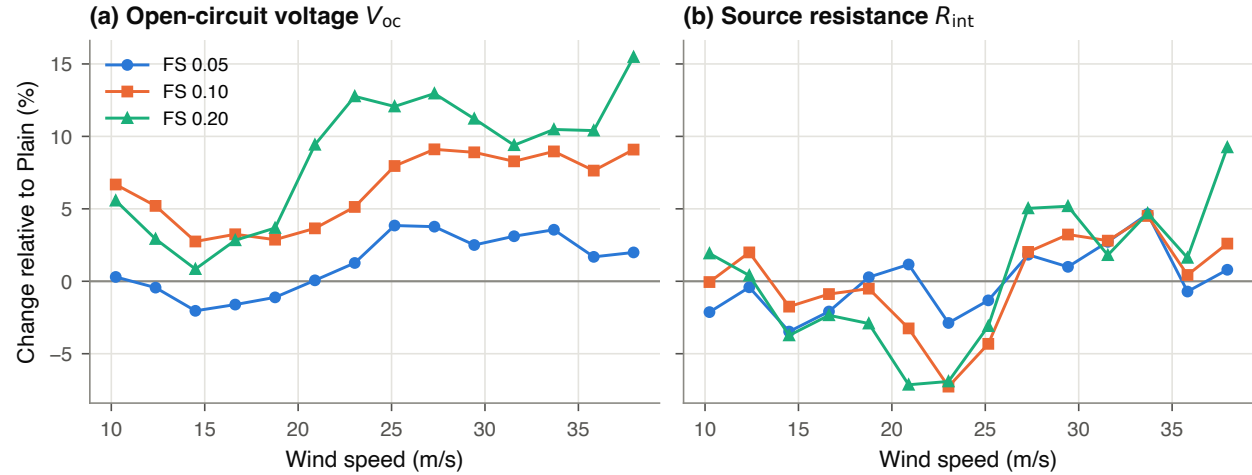


Figure 7. Thévenin decomposition relative to the Plain rotor (geometric mean of two runs). (a) Extrapolated open-circuit voltage. (b) Apparent source resistance. All curves share the Plain reference, so features common to all three may come from it.

period, so their difference (+10.8% [+8.2, +13.5]) is the comparison least affected by layer height.

- **Run order.** The rotors were run in order of increasing texture, so slow drift is confounded with texture. A fit of the eight run means with a rotor term and a linear time term gives a drift of +0.8% per hour (95% interval −1.7% to +3.3%). With that term, FS 0.10 and FS 0.20 remain resolved from Plain (+11.5% [+7.0, +16.1] and +14.5% [+8.9, +20.4]; t intervals, 3 degrees of freedom), while FS 0.05 is +0.9% [−2.3, +4.2]. A linear drift of this size cannot explain the FS 0.10 and FS 0.20 differences but could account for the FS 0.05 difference; a change that occurred between rotors, rather than steadily, cannot be separated from a rotor effect with these data.

8 Conclusions

1. On 1 October 2026 the FS 0.10 and FS 0.20 rotors produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak electrical power than the Plain rotor, averaged over 10.2–38.0 m/s. The FS 0.05 difference, +1.8% [−0.6, +4.2], was not resolved and could be produced by session drift.
2. The advantage varies with wind speed: the power coefficient of FS 0.10 and FS 0.20 rises more steeply, and at lower wind speed, than Plain's, so the gains are largest at 21–25 m/s.
3. Averaged over wind speed, the gain is carried by the extrapolated open-circuit voltage, which indicates faster rotation of the textured rotors at light load; where the gains are largest, a lower apparent source resistance also contributes.
4. R_a and P_a , as scanned (one field per set), do not order the rotors by power: FS 0.05 and FS 0.10 have similar P_a but differ by +10.8% in power, so the scans do not confirm that their realised texture differs.
5. The intervals reflect run-to-run repeatability on one print and one mounting per texture. The FS 0.10 and FS 0.20 differences from Plain would stay resolved with mounting variation of up to about 2.0–2.6% (Tukey criterion); replicate prints and mountings are needed to measure that variation and to attribute the differences to texture.

Table 6. Sources of uncertainty.

Source	Size	Effect on the rotor comparison
Run-to-run repeatability	$\sigma = 0.58\%$ (run mean); 1.9% per point	In every interval.
Peak estimator	Changes within 0.5 percentage points	FS 0.20 vs FS 0.10 becomes marginal (Table 4).
Drift during the session	+0.8%/h $[-1.7, +3.3]$	Confounded with run order; a linear-drift fit leaves FS 0.10 and FS 0.20 resolved.
Ramp dynamics	1 s dwells, equilibrium not verified; at 500 rpm the voltage rose between the first two dwells in seven of eight runs	Common protocol, but a bias that depends on rotor inertia or torque-speed slope would not cancel.
Wind-speed calibration	2.8–4.6% high at 10–14 m/s; 5% above a line through the measured points only at 38.0 m/s (an extrapolation, not a bound)	None on the comparison (identical fan speeds); affects absolute v and $C_{P,el}$.
Air density	Not recorded	Any change is confounded with run order (see drift); $C_{P,el}$ assumes standard air.
Blockage	Not corrected	Similar for rotors of one geometry; not quantified (test-section size not documented).
Electrical measurement	Not quantified	Largely cancels (same instrument and settings).
Print and mounting	Not measured	Not in the intervals; Section 5 gives the variation each difference can absorb.

9 Recommendations

1. **Replicate and randomise.** Print at least two sets per thickness, including a Plain set at the textured sets' 0.1 mm layer height. Mount each set twice and run in an interleaved order, re-running the Plain reference after every third rotor. This measures print and mounting variation and removes the run-order confound.
2. **Add rotor speed.** Join T. Kang's rotor-speed record to the dwells to obtain tip-speed ratio λ and $C_{P,el}(\lambda)$. Separating aerodynamic from parasitic torque also needs a spin-down or motored friction test of each mounted rotor.
3. **Extend the wind calibration** with a direct air-speed measurement over the test range; record air temperature and pressure on every run; document the test-section dimensions for a blockage estimate.
4. **Characterise the surfaces at fixed stations:** several fields per set on two blades and both faces, profiles along and across the layer lines, and an evaluation length of at least five cut-off lengths. Keep the sliced file, printer, nozzle and filament record for every print.

9.1 Toward the fault-testing plan

Prof. Jeong asked for a fault-testing plan agreed before the next round. The rig offers an automated sweep of about 8 minutes with run-to-run repeatability of 0.58% (run mean; repeat runs on one mounting, back to back except Plain's, 60 minutes apart). Whether V_{oc} and R_{int} separate rotor-side from electrical faults is untested: R_{int} also reflects the rotor slowing under load (Section 5), so this

should be checked on the first fault cases together with rotor speed. Candidate faults are a tip mass on one blade (imbalance); a chipped, shortened or missing blade; added bearing drag; and an open or high-resistance generator phase. A short meeting to agree the fault list, severity levels and acceptance criteria should precede any fault prints.

Acknowledgement

T. Kang provided the electronic load and recorded rotor speed during the session.

References

- [1] ISO 16610-21:2011, *Geometrical product specifications (GPS) — Filtration — Part 21: Linear profile filters: Gaussian filters*.
- [2] D. C. Montgomery, *Design and Analysis of Experiments*, 9th ed., Wiley, 2017.
- [3] ISO 4288:1996, *Geometrical product specifications (GPS) — Surface texture: Profile method — Rules and procedures for the assessment of surface texture*; superseded by ISO 21920-3:2021.

A Peak power by run and set point

Table 7. Peak electrical power $P_{\max}$ (W), largest dwell, every run. Runs 1 and 2 share the rotor’s mounting.

Fan rpm	v m/s	Plain		FS 0.05		FS 0.10		FS 0.20	
		1	2	1	2	1	2	1	2
500	10.2	0.0294	0.0298	0.0313	0.0317	0.0330	0.0348	0.0303	0.0339
600	12.4	0.0658	0.0660	0.0638	0.0633	0.0707	0.0704	0.0676	0.0660
700	14.5	0.1140	0.1175	0.1128	0.1153	0.1197	0.1184	0.1232	0.1218
800	16.6	0.1858	0.1811	0.1808	0.1794	0.1931	0.1969	0.1983	0.2023
900	18.8	0.2753	0.2937	0.2864	0.2644	0.3045	0.3135	0.3181	0.3220
1000	20.9	0.4162	0.4109	0.4158	0.4044	0.4716	0.4671	0.5362	0.5516
1100	23.0	0.6093	0.6031	0.6474	0.6572	0.7309	0.7381	0.8278	0.8240
1200	25.2	0.8894	0.8914	0.9700	0.9642	1.0875	1.0921	1.1261	1.1142
1300	27.3	1.2706	1.2463	1.3247	1.2875	1.4130	1.4123	1.4403	1.4474
1400	29.4	1.5788	1.5975	1.6391	1.6184	1.7898	1.7959	1.8351	1.8511
1500	31.6	1.9799	2.0368	2.0413	2.0236	2.2892	2.2066	2.2762	2.2513
1600	33.7	2.4530	2.4832	2.4978	2.5259	2.7952	2.8414	2.9085	2.8549
1700	35.8	3.0003	3.0273	3.0671	3.0700	3.4574	3.5213	3.5436	3.5516
1800	38.0	3.6640	3.6666	3.7998	3.7517	4.3137	4.2905	4.4798	4.5882

B Statistical details

Model. With $y_{rjs} = \ln P_{\max}$ for rotor r (4), run j (2) and set point s (14),

$$y_{rjs} = \mu + \alpha_r + \beta_s + (\alpha\beta)_{rs} + u_{rj} + e_{rjs},$$

where u_{rj} is a run offset within the rotor and e_{rjs} the residual [2]. The rotor effect is tested against run within rotor, which is the one-way ANOVA on run means. The interaction $(\alpha\beta)_{rs}$ and the run

term are tested against the residual (Table 8). Run offsets do not enter either mean square, so the interaction test needs no assumption about them; it does assume equal residual variance, so it is approximate here.

Table 8. Analysis of variance of $\ln P_{\max}$. MS in units of 10^{-4} .

Source	df	MS	F	p	Tested against
Rotor	3	1 550	324	$p < 0.001$	run within rotor
Set point (wind speed)	13	187 000			
Rotor $\times$ set point	39	31.7	9.22	$p < 0.001$	residual
Run within rotor	4	4.78	1.39	$p = 0.25$	residual
Residual	52	3.44			

Comparisons. With run means $\bar{y}_{rj}$, rotor means $\bar{y}_r$ and pooled $\sigma^2 = \sum_r \sum_j (\bar{y}_{rj} - \bar{y}_r)^2 / 4$, Tukey’s interval for rotors a and b is $\exp(\bar{y}_b - \bar{y}_a \pm q_{0.05;4,4} \sigma / \sqrt{2}) - 1$, a half-width of 0.024 in $\ln P_{\max}$ (about 2.4%). The same procedure is applied to $\ln V_{oc}$ and $\ln R_{int}$. The permutation test enumerates all 105 ways of splitting the eight run means into four pairs.

Mounting variation. If each rotor’s mean carried an independent mounting offset with standard deviation τ , the difference of two rotor means would have variance $\sigma^2 + 2\tau^2$. It stays resolved by the Tukey criterion while $|\bar{y}_b - \bar{y}_a| > (q_{0.05;4,4} / \sqrt{2}) \sqrt{\sigma^2 + 2\tau^2}$, which gives the values of τ quoted in Section 5.

Drift. The run means are fitted as $\bar{y}_{rj} = a_r + \beta t_{rj}$, with t_{rj} the start time of the run in hours, by least squares (3 residual degrees of freedom).

Residual scatter. The per-point residual standard deviation is 1.9% overall; it is largest at 500 rpm (4.1%) and smallest at 1400 rpm (0.2%).

C Surface analysis

The Keyence export is a height map in millimetres, quantised to $1 \mu\text{m}$. Each image column is one spanwise profile. A column is dropped if more than 10% of its points are missing, or if more than 2% are missing between its first and last valid points (1.8% of columns or fewer for Plain, FS 0.05 and FS 0.10; 5.1% for FS 0.20, where missing points concentrate on steep slopes, so its values may be slightly low). The remaining interior gaps are filled linearly, a least-squares line is removed, and the profiles of a field are cut to the length of the shortest one. P_a is the mean absolute deviation of the levelled profile. For R_a the levelled profile minus its ISO 16610-21 Gaussian mean line ($\alpha = \sqrt{\ln 2 / \pi}$, $\lambda_c = 0.25 \text{ mm}$) is evaluated with $\lambda_c / 2$ discarded at each end [1]. The layer period is the peak of the mean power spectrum between 60 and $320 \mu\text{m}$. For periodic profiles ISO 4288 [3] selects $\lambda_c = 0.25 \text{ mm}$ for spacings of 0.04–0.13 mm, which fits FS 0.05 and FS 0.10. The Plain set’s 0.2 mm period would call for 0.8 mm, and the non-periodic FS 0.20 surface (R_a above $10 \mu\text{m}$) for 8 mm, both longer than the 1.4 mm field. The filter transmits 50% at λ_c : R_a keeps about 65% of the amplitude of the Plain set’s 0.20 mm layer lines and 99% of the 0.10 mm ones, so Plain’s R_a is understated relative to FS 0.05 and FS 0.10. P_a is unfiltered.

D Wind-speed calibration

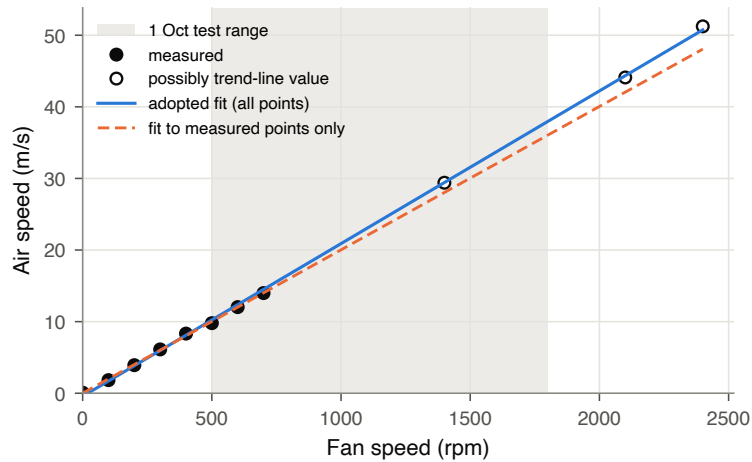


Figure 8. Tunnel calibration, Test 1 (13 February 2026). Filled: measured; hollow: marked in the calibration file as possibly read from the trend line of the original record. Shaded: the range of this test.

Table 9. Calibration points and the adopted fit $v = 0.02132N - 0.424$ (Fig. 8). A line through the measured points alone has residual standard deviation 0.19 m/s.

Fan (rpm)	v (m/s)	Source	Fit (m/s)	Fit – measured (m/s)
0	0.08	measured	−0.42	−0.50
100	1.83	measured	1.71	−0.12
200	3.92	measured	3.84	−0.08
300	6.12	measured	5.97	−0.15
400	8.33	measured	8.10	−0.23
500	9.79	measured	10.24	+0.45
600	12.03	measured	12.37	+0.34
700	13.99	measured	14.50	+0.51
1400	29.40	possibly trend line	29.43	+0.03
2100	44.10	possibly trend line	44.35	+0.25
2400	51.24	possibly trend line	50.75	−0.49

E Data package and reproducibility

The accompanying archive contains:

- `1_rig_sweeps/`: the raw files of the eight runs;
- `2_surface_scans/`: the profilometer height maps;
- `3_derived/`: the analysis-ready tables behind this report;
- `4_reference/`: the rotor geometry, blade mesh, wind calibration and slicer summary;
- `5_figures/` and `6_code/`: the figures and the analysis code.

`README.md` explains the layout, `DATA_DICTIONARY.md` defines every column, and `MANIFEST.csv` gives each file’s SHA-256 checksum. The script `6_code/build_report.py` (Python 3) regenerates every result, table and data figure in this report from the raw files.